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Parametrization-free locally-conformal perfectly matched layer method for finite element solution of Helmholtz equation ☆

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A B S T R A C T

We present a novel Locally-Conformal Perfectly Matched Layer (LCPML) method for mesh truncation in the Finite Element Method (FEM), to solve wave equations derived from Maxwell's equations. The original LCPML method was proposed by Ozgun and Kuzuoglu in 2006 [16] by utilizing the concept of complex elements in which nodal coordinates of elements are replaced by complex coordinates obtained by using a complex coordinate-transformation. In the proposed LCPML method, named as LCPML-log method, a more efficient coordinate-transformation is used together with new FEM formulation and implementation. The proposed method offers the advantage that only a few (or even a single) PML layer is sufficient to get reliable results. In addition, it does not require any PML parameter tuning, and hence, the decaying wave behavior in PML is achieved self-adaptively. The numerical results of some detailed and systematic performance analyses are also presented.

1. Introduction

Perfectly Matched Layer (PML) is an artificial absorbing layer that is commonly used to truncate the computational region in *finite* methods for solving Maxwell's equations in an unbounded region, using the Finite Element Method (FEM) or the Finite Difference Time Domain (FDTD) methods. The PML is a preferred choice over classical absorbing boundary conditions (ABCs), because of its simplicity, versatility and efficiency. Its key feature is that it can be designed to be conformal and its inner boundary can be very close to sources. This, in turn, serves to minimize the white space of the computational domain, even as it absorbs the outgoing waves incident upon the PML region without introducing artificial reflections back from the truncation boundary into the interior of the PML region.

The PML concept was originally introduced by Berenger [1] here in the context of the FDTD method for solving Maxwell's equations. The PML, proposed by Berenger, splits the electromagnetic fields inside the PML region into two non-Maxwellian fields, and hence, it is referred to as the *split-field PML*. An alternative PML formulation for the FDTD, called the *uniaxial PML*, has been proposed by Gedney [2], in which the PML is an artificial anisotropic medium. A more general approach, which was introduced by Chew and Weedon [3], also for the FDTD, is known as *stretched-coordinate PML* because it is based on a complex coordinate transformation that produces non-Maxwellian fields within the PML region. Subsequently, an alternative PML in FEM modeling was introduced by Sacks *et al.* in [4], where the authors designed a Maxwellian PML as an *anisotropic layer* with artificial material tensors defined by using a complex coordinate-stretching. First introduced in Cartesian coordinates, this approach was later extended to cylindrical and spherical coordinates to design conformal PMLs using a locally curvilinear coordinate system, first by Kuzuoglu and Mittra in [5,6] and later by Teixeira and Chew in [7]. Although the PML, introduced in the 1990s, was first used to solve Maxwell's equations for electromagnetic waves, it was successfully applied later to other disciplines, *e.g.*, acoustics [8–11], elastodynamics [12,13] and the solution of linearized Euler's equations [14,15].

Subsequently, in 2006, Ozgun and Kuzuoglu introduced a highly flexible and accurate mesh truncation scheme, called the *Locally-Conformal PML* (LCPML) method [16–19], for mesh truncation of arbitrarily shaped computational domains in FEM modeling. In this

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method, the attenuation of waves within the PML region is achieved by defining a *polynomial* decay function that absorbs waves arriving in the PML region without artificial reflections from the PML boundary. Using this decay function, a locally-defined complex coordinate transformation is defined and the method is simply implemented by just replacing the real coordinates by complex coordinates, whose real parts are the original coordinates, but whose imaginary parts are obtained by using a special coordinate transformation designed to achieve the desired effects of PML action. Specifically, the method utilizes the concept of *complex elements* where the nodal coordinates of elements are complex- rather than real-valued, and hence, the LCPML method can be easily implemented in an FEM code without modifying the conventional formulation. Extensions of the LCPML have also been proposed in the literature. In [10], Beriot et al. have introduced the Automatically Matched Layer (AML), in which the mesh of the layer is extruded automatically and the practical implementation relies on a simple modification of the Jacobian matrix in the element wise integrals. The LCPML and AML formulations have been both recently applied to Helmholtz problems discretized using Isogeometric Analysis [11].

In this paper, we propose a new LCPML method, which differs from the previous method, *not only* by how the decay function is defined using a different coordinate transformation, *but also* by how the method is implemented in an FEM code. The authors employ a *logarithmic* decay function and the effect of the coordinate transformation is integrated into the FEM by using a new formalism where the Jacobian matrices are computed via direct transformation, bypassing the use of the complex elements altogether. The method is named as LCPML-log, where 'log' refers to the logarithmic decay function. A salutary feature of the LCPML-log method is that a few (e.g., 1 to 3) PML layers are sufficient to get reliable results, since the reduction of the number of PML layers obviously decreases the runtime. Another advantage of the proposed method is that it does not require any special parameter setting for PML design. Furthermore, compared to the AML formulation from [10], the proposed approach is more general, as it also allows to tackle PML that is not extruded, but rather generated from conventional meshing procedures (e.g. through triangular filling).

The paper is organized as follows: In Sec. 2, we introduce the PML concept with complex coordinate stretching for the Helmholtz equation. In Secs. 3 and 4, coordinate transformations and FEM formulations, respectively, which are used in both original LCPML and new LCPML-log methods are described in detail. In Sec. 5, the results of several numerical simulations are presented to show the performance of the LCPML-log method. Finally, in Sec. 6, some conclusions are drawn.

2. PML concept with complex coordinate stretching for the Helmholtz equation

A general time-harmonic electromagnetic wave propagation problem (either radiation or scattering problem) in an unbounded medium $\mathbf{\Omega}_\infty$ in $\Re^2$ is governed by the Helmholtz equation as follows (assuming a suppressed time-dependence $\exp(j\omega t)$):

$$\nabla \cdot (p\nabla u) + qu = f_e(\mathbf{r}), \tag{1a}$$

which is explicitly written as

$$\frac{\partial}{\partial x}\left(p\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(p\frac{\partial u}{\partial y}\right) + qu = f_e(\mathbf{r}), \tag{1b}$$

where $\mathbf{r} \in \mathbf{\Omega}_\infty$; the unknown function $u(\mathbf{r})$ is $E_z(\mathbf{r})$ and $H_z(\mathbf{r})$ in $\mathrm{TM_z}$ (transverse magnetic) and $\mathrm{TE_z}$ (transverse electric) polarizations, respectively; and $f_e(\mathbf{r})$ is the excitation/source function. In addition, $p = \mu_r^{-1}$ and $q = k^2\varepsilon_r^{-1}$ in $\mathrm{TM_z}$ polarization, and $p = \varepsilon_r^{-1}$ and $q = k^2\mu_r^{-1}$ in $\mathrm{TE_z}$ polarization, where k is the free-space wavenumber, and ε_r and μ_r are the relative permittivity and permeability, respectively, of the medium. Since the spatial domain is unbounded, asymptotic behavior of the waves radiated from localized sources is governed by the Sommerfeld radiation condition at infinity, as follows:

$$\lim_{\rho\to\infty} \sqrt{\rho}\left(\frac{\partial u}{\partial \rho} + jku\right) = 0 \tag{2}$$

where ρ is the distance from sources.

Since the spatial domain of interest extends to infinity, the radiation condition must be *explicitly* imposed in finite methods (e.g., FEM or FDTD) since the spatial domain must be of finite size in such methods. A general approach to mimic the effect of the radiation condition in finite methods is to truncate the unbounded domain by introducing an artificial boundary or layer defined around a region containing all sources, so that outgoing waves arriving at the truncation boundary are absorbed without being reflected from the boundary. Among various mesh truncation approaches, Perfectly Matched Layers (PMLs) are very powerful in terms of the ease of use and higher accuracy with less computational load. Although there are different PML implementations in the literature, one elegant and general approach is the *stretched-coordinate PML* approach based on complex coordinate stretching. The general idea in a stretched-coordinate PML is to use a coordinate transformation in which the real coordinates inside the PML region are mapped to complex coordinates. Therefore, the wave propagation problem in an unbounded medium $\mathbf{\Omega}_\infty$, which is governed by the Helmholtz equation in (1) together with the Sommerfeld radiation condition in (2), can be formulated as a boundary value problem defined in a bounded medium $\mathbf{\Omega}$ as follows:

$$\frac{\partial}{\partial x}\left(p\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial y}\left(p\frac{\partial u}{\partial y}\right) + qu = f_e(\mathbf{r}), \quad \text{for } \mathbf{r} \in \mathbf{\Omega}\backslash\mathbf{\Omega}_{\mathrm{PML}} \tag{3a}$$

$$\frac{\partial}{\partial \tilde{x}}\left(p\frac{\partial u}{\partial \tilde{x}}\right) + \frac{\partial}{\partial \tilde{y}}\left(p\frac{\partial u}{\partial \tilde{y}}\right) + qu = 0, \quad \text{for } \tilde{\mathbf{r}} \in \mathbf{\Omega}_{\mathrm{PML}} \tag{3b}$$

where $\mathbf{r} \in \Re^2$ and $\tilde{\mathbf{r}} \in \mathbb{C}^2$ are the coordinates in real and complex spaces, respectively.

There are different stretched-coordinate PML approaches in the literature based on the definition of the coordinate transformation and the implementation details. The *classical PML* method was proposed in the literature to design PMLs over rectangular and circular regions using two-dimensional Cartesian and cylindrical coordinate systems [5,6]. Let us first define the classical PML method by considering a

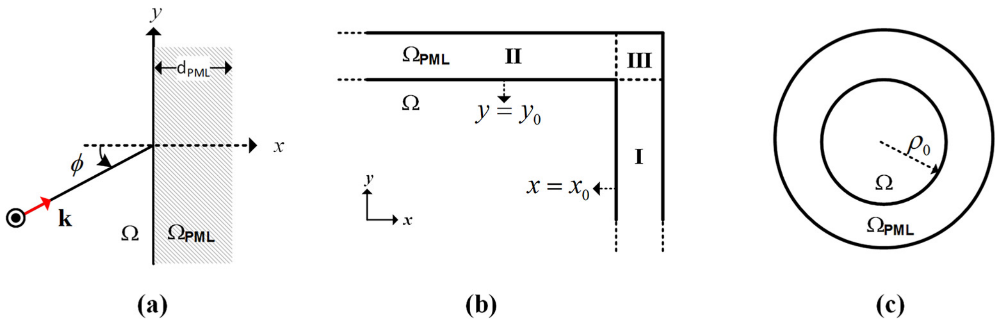

Fig. 1. Illustration of the classical PML method. (a) Along x-axis, (b) in rectangular region, (c) in circular region.

plane wave incident to a planar boundary at $x = 0$ as shown in Fig. 1(a), where the half-space $x > 0$ is the PML region. The plane wave in $x < 0$ and $x > 0$ regions is expressed as follows:

$$u = u_0 e^{-jk(x\cos\phi + y\sin\phi)} \quad \text{in } \mathbf{\Omega}(x < 0) \tag{4}$$

$$u = u_0 e^{-jk(x\cos\phi + y\sin\phi)} e^{-\alpha x\cos\phi} = u_0 e^{-jk\tilde{x}\cos\phi} e^{-jky\sin\phi} \quad \text{in } \mathbf{\Omega}_{\mathrm{PML}}(x > 0) \tag{5a}$$

where $\tilde{x}$ is the complex coordinate given as

$$\tilde{x} = x\left(1 + \frac{\alpha}{jk}\right) \tag{5b}$$

In (5a), the plane wave in the PML region is damped along the x-direction by an exponentially decaying function where $\alpha > 0$.

This approach can be generalized to a rectangular region as follows (see Fig. 1(b)):

$$\begin{array}{lll} \text{In Region I:} & \text{In Region II:} & \text{In Region III:} \\ \tilde{x} = x + \dfrac{\alpha}{jk}(x - x_0) & \tilde{x} = x & \tilde{x} = x + \dfrac{\alpha}{jk}(x - x_0) \\ \tilde{y} = y & \tilde{y} = y + \dfrac{\alpha}{jk}(y - y_0) & \tilde{y} = y + \dfrac{\alpha}{jk}(y - y_0) \end{array} \tag{6}$$

Similarly, the classical PML method for the circular PML domain is expressed in cylindrical coordinates as (see Fig. 1(c))

$$\tilde{\rho} = \rho + \frac{\alpha}{jk}(\rho - \rho_0) \tag{7}$$

Two main drawbacks in the classical PML method are as follows: (i) The inner and outer PML surfaces must be defined on *constant coordinate surfaces*, and hence, the classical PML can only be used for canonical (e.g., rectangular, circular) geometries, and hence, arbitrary geometries cannot be handled. (ii) The performance of the classical PML method depends on the choice of parameters (α and d_{PML}) defined in the transformation. The value of α must be specified to achieve smooth decay of the transmitted wave in the PML region. In addition, d_{PML} is chosen such that the term $e^{-\alpha x\cos\phi}\big|_{x=d_{\mathrm{PML}}}$ decays to a negligible value. Although large α values meet the last requirement, they may cause deformations in the transformed coordinates due to large imaginary parts. Thus, an optimal value of α is essential for a successful PML design. Typically, the parameter α is determined as $\alpha = \alpha_c k$, where k is the wavenumber and α_c is a positive real number. The performance of the classical PML as a function of α_c at different frequencies will be numerically illustrated in Sec. 5. The classical PML starts to lose its efficiency when the number of PML layers is relatively small.

In [16–18], Ozgun and Kuzuoglu proposed the *locally-conformal PML* (LCPML) method, which is a very flexible and easy-to-implement method based on a complex coordinate transformation specially-defined to design conformal PML over an *arbitrary* convex domain. Unlike the classical PML, the LCPML method does not require any constant coordinate surface, making it easily applicable to PML regions of arbitrary shape. Its ease of implementation is due to the fact that the LCPML method does not require additional geometric mesh data and uses the standard FEM formulation without any modification except that the real coordinates within the PML region are replaced by complex coordinates using a specially- and locally-defined complex coordinate transformation which adds an imaginary part to each real coordinate. The imaginary part of the coordinate transformation replaces propagating waves within a PML region by damped waves with exponentially decaying envelope. This is essentially analogous to damped oscillating waves propagating in a lossy medium. In the LCPML method, the complex coordinates are represented as

$$\tilde{\mathbf{r}} = \mathbf{r} + \frac{1}{jk} f(\delta)\,\hat{\mathbf{n}}(\delta) \tag{8}$$

The imaginary part of the complex coordinate, which is $f(\delta)\,\hat{\mathbf{n}}(\delta)/k$, is obtained by a coordinate transformation, which will be defined in Sec. 3 in the context of the original and proposed LCPML methods. Here, $f(\delta)$ is called the *decay function*, which is a monotonically increasing function of the *decay parameter* δ, and $\hat{\mathbf{n}}$ is a unit vector representing the direction of the decay within the PML region. When the real coordinate $\mathbf{r}$ is replaced by the complex coordinate $\tilde{\mathbf{r}}$, any outgoing plane wave traveling along the propagation unit vector $\hat{\mathbf{k}}$ experiences absorption in the direction of the unit vector $\hat{\mathbf{n}}$ as follows:

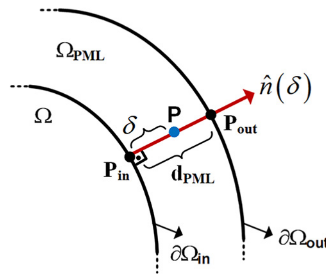

Fig. 2. Illustration of the LCPML parameters.

$$u = u_0 \exp\left(-jk\hat{\mathbf{k}}\cdot\tilde{\mathbf{r}}\right) = u_0 \exp\left(-jk\hat{\mathbf{k}}\cdot\mathbf{r}\right)\exp\left(-f(\delta)\,\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}\right) \tag{9}$$

where $\exp\left(-f(\delta)\,\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}\right)$ is an exponentially decaying envelope function, which annihilates the oscillatory waves within the PML region in a specified direction. It is clear that the waves propagating along the unit vector are most attenuated as the term $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}}$ takes on its maximum value (i.e., when $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}} = 1$). Hence, the choice of the decay function and the unit vector is important for efficient PML design, which will be described in Sec. 3.

3. Coordinate transformations of the original LCPML and the proposed LCPML-log methods

In the LCPML method, the general form of the complex coordinate transformation is given in (8), where an imaginary part is added to each real coordinate in the PML region. In this transformation, the PML region $\mathbf{\Omega}_{PML}$, which is originally a subset of $\Re^2$, is extended to $\mathbf{\Gamma} \subset \mathbb{C}^2$, which is a manifold in $\mathbb{C}^2$. This approach does not require any locally-defined coordinate system, and each point P in $\mathbf{\Omega}_{PML}$ defined in the original coordinate system is mapped to $\tilde{\mathrm{P}}$ in $\mathbf{\Gamma}$. In the following subsections, the main ingredients of the coordinate transformation (i.e., decay function, decay parameter and unit vector) are described for the original LCPML and the proposed LCPML-log methods.

3.1. Original LCPML method [16–18]

The method is illustrated in Fig. 2. The decay parameter $\delta = [0, 1]$ is defined as follows:

$$\delta = \frac{\|\mathbf{r} - \mathbf{r}_{in}\|}{d_{PML}} = \frac{\left[(x - x_{in})^2 + (y - y_{in})^2\right]^{1/2}}{d_{PML}}, \tag{10}$$

where d_{PML} is the local PML thickness defined as

$$d_{PML} = \|\mathbf{r}_{out} - \mathbf{r}_{in}\| = \left[(x_{out} - x_{in})^2 + (y_{out} - y_{in})^2\right]^{1/2} \tag{11}$$

where $\mathbf{r} \in \mathbf{\Omega}_{PML}$ is the position vector of the PML point $\mathrm{P} = (x, y)$, as well as $\mathbf{r}_{in} \in \partial\mathbf{\Omega}_{in}$ and $\mathbf{r}_{out} \in \partial\mathbf{\Omega}_{out}$ are the position vectors of the points $\mathrm{P}_{in} = (x_{in}, y_{in})$ and $\mathrm{P}_{out} = (x_{out}, y_{out})$ located on the inner and outer PML boundaries, respectively. Furthermore, $\|.\|$ represents the Euclidian norm, so it is related to the distance between the corresponding points. The point P_{in} is selected as the boundary point closest to the PML point P. Hence, it is the solution of the minimization problem: $\min_{\mathbf{r}_{in} \in \partial\mathbf{\Omega}_{in}} \|\mathbf{r} - \mathbf{r}_{in}\|$, which results in a unique $\mathbf{r}_{in}$ since the boundary $\partial\mathbf{\Omega}_{in}$ is a convex set. The point P_{out} is the point on the outer PML boundary, which is determined as the intersection of the line along the unit vector $\hat{\mathbf{n}}(\delta)$ with the outer boundary. The unit vector is *the normal unit vector* passing through P and P_{in}, which is defined as

$$\hat{\mathbf{n}}(\delta) = \frac{\mathbf{r} - \mathbf{r}_{in}}{\|\mathbf{r} - \mathbf{r}_{in}\|} \tag{12}$$

The direction of the unit vector along the normal direction indicates the damping direction of the wave within the PML region. The decay function is defined as a monotonically increasing function of the decay parameter δ, as follows:

$$f(\delta) = \frac{\alpha}{m}\delta^m d_{PML} \tag{13}$$

where $\alpha = \alpha_c k$ (α_c is a positive real number) and m is a positive integer (typically 2 or 3) related to the damping rate of the wave inside the PML region. These parameters must be specified in such a way that the transmitted wave within the PML region should be absorbed monotonically and the magnitude of the wave must be negligible when the wave reaches the outer PML boundary. The choice of the parameters (α and m) was discussed in [16–18] in detail. We had suggested in these studies that, to get reliable results, the parameter α_c should be in the range $5 \le \alpha_c \le 10$ for the PML thickness between $\lambda/4$ and $\lambda/2$. However, similar to the classical PML method, the accuracy may decrease for a fixed α_c parameter especially for a frequency sweep over a broader band of frequencies.

Since the performance of the LCPML method depends on the α_c parameter, we propose in this study an alternative approach. The main action of this transformation is to multiply the plane wave expression by an exponentially decaying envelope function as shown in (9). Let us consider the case where the wave propagates along the unit vector (i.e., $\hat{\mathbf{k}}\cdot\hat{\mathbf{n}} = 1$) in (9) without loss of generality. Hence, the magnitude of the wave within the PML region is damped by the real exponential term $\exp(-f(\delta))$. The exponential envelope function must attain a negligible value to annihilate the wave on the outer PML boundary (i.e., when $\delta = 1$) because otherwise artificial reflections will occur from this boundary. That is, the following condition

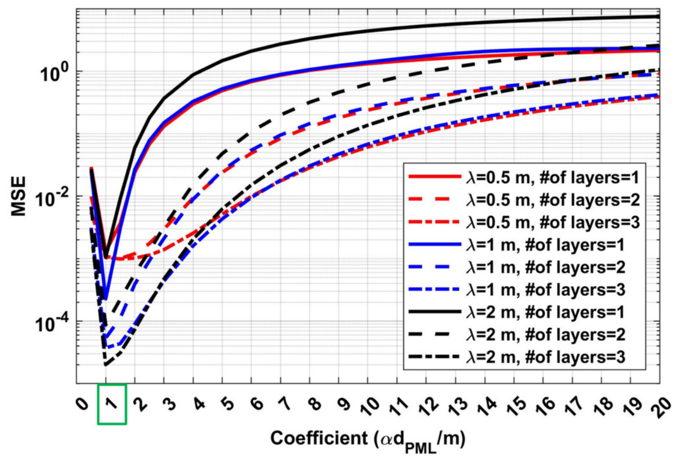

Fig. 3. MSE versus $\alpha d_{\mathrm{PML}}/m$ parameter of the general form in (17) to show that the special form in (18) is the optimal choice (*y-axis is logarithmic*). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\exp\left(-f\left(\delta\right)\right)\big|_{\delta=1} \approx \epsilon \tag{14}$$

should be satisfied, where ϵ is a small number close to zero, e.g., $\epsilon = 10^{-2}$. By substituting the decay function given in (13) into (14), the parameter α can be expressed as follows:

$$\alpha = -\log\left(\epsilon\right)\frac{m}{d_{\mathrm{PML}}} \tag{15}$$

which simply makes the decay function as

$$f\left(\delta\right) = -\log\left(\epsilon\right)\delta^{m} \tag{16}$$

Note that this approach is still not parameter-free as the decay function depends on the choice of m and ϵ.

3.2. Proposed LCPML-log method

In the LCPML-log method, let us first propose two forms (general and special forms) of the decay function as follows:

$$\text{General form: } f\left(\delta\right) = -\frac{\alpha}{m}d_{\mathrm{PML}}\log\left(1-\delta\right) \tag{17}$$

$$\text{Special form: } f\left(\delta\right) = -\log\left(1-\delta\right) \tag{18}$$

where α and m can be interpreted as the same parameters in the original LCPML. However, when we have tested the general form with a different set of α and m parameters, we have observed that the optimal solution has been achieved when $\alpha d_{\mathrm{PML}}/m = 1$ and $m = 1$, which corresponds to the special form in (18). Some of the results from our test simulations are demonstrated in Fig. 3, where a circular domain with radius $a = 1$ m is considered with different number of PML layers and different wavelength (λ) values, similar to the simulations described in Sec. 5. In this graph, the MSE (relative mean-square error) values, defined in (60), are plotted as a function of the coefficient $\alpha d_{\mathrm{PML}}/m$. It is clear from the results that the optimal solution is obtained when the coefficient in front of the log function is unity, which means that the parameter-free special form in (18) is the best choice for a PML design. This best choice was also verified by Bermudez et al. [9] such that this specific choice of stretching function that does not enforce the continuity condition for the derivative of the decay function provides better accuracy compared to the other choices shown on page 477 of [9].

An important ingredient of the coordinate transformation is the unit vector, which is defined as the 'normal' unit vector as in the original LCPML method, because of the outgoing behavior of the waves away from sources. To enable the normal unit vector as the absorption direction in the proposed FEM formulation, the derivative terms in Jacobian matrices are computed 'numerically', as will be described in Sec. 4.

This logarithmic decay function, originally introduced by Bermudez et al. in [9], has an advantage and a disadvantage compared with the polynomial decay function used in the original LCPML method formulated in (13) or (16). As will be demonstrated in Sec. 4, the logarithmic decay function outperforms the polynomial decay function especially when a few (1 to 3) PML layers are used. Therefore, the decay function in (18) is advantageous in terms of decreasing the number of unknowns used in the PML region. The attenuation of a propagating wave within a PML region is illustrated in Fig. 4 for both logarithmic and polynomial ($m = 2$) decay functions given in (18) and (16), respectively. In these plots, the envelope function represents the real exponential function $\exp\left(-f\left(\delta\right)\right)$. It is useful to note that the effect of the logarithmic decay function over a propagating wave is linear as $\exp\left(-f\left(\delta\right)\right) = 1-\delta$. When linear basis functions are used in the FEM, which are widely used due to their simplicity and efficiency, the linear behavior of the envelope of the logarithmic decay function is well suited to the linear behavior of the basis functions, thus increasing the accuracy. As for the disadvantage of the logarithmic decay function in (18), it suffers from logarithmic singularity as $\lim_{\delta\to 1} f\left(\delta\right) \to \infty$. In other words, the coordinate transformation fails due to this singularity when a PML point is 'exactly' on the outer PML boundary. Therefore, the simple idea of replacing the real coordinates by complex coordinates, as used in the original LCPML method, is not employed in the proposed LCPML-log method. To avoid the direct use of coordinates lying on the outer PML boundary, a new FEM formulation is proposed, which is explained in Sec. 4.

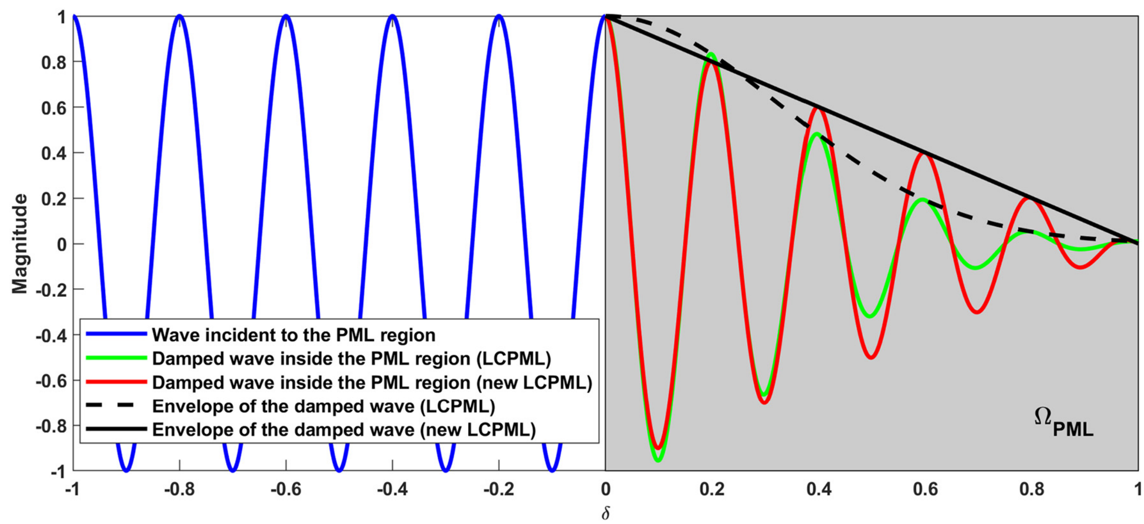

Fig. 4. Illustration of the wave behavior within a PML region.

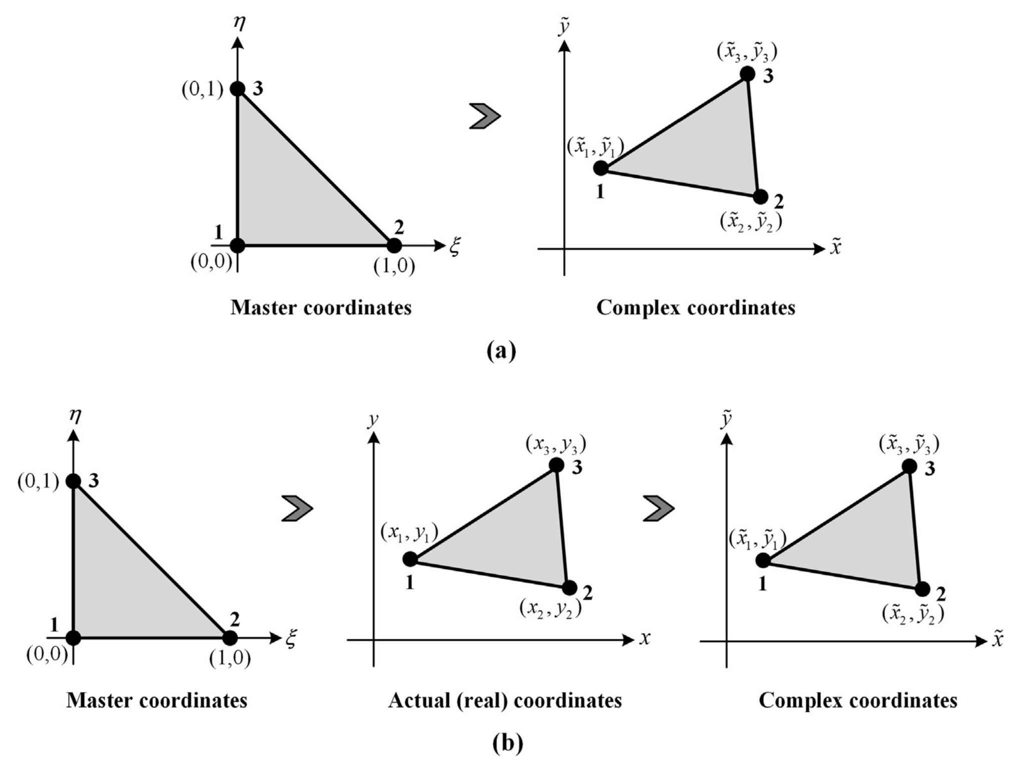

Fig. 5. Transformation between complex and master coordinates. (a) $(\xi,\eta)\to(\tilde{x},\tilde{y})$ used in the original LCPML method, (b) $(\xi,\eta)\to(x,y)\to(\tilde{x},\tilde{y})$ used in the LCPML-log method.

4. FEM formulations of the original LCPML and the proposed LCPML-log methods

As discussed in Sec. 2, the Helmholtz equation in the PML region is expressed in terms of complex coordinates as follows:

$$\tilde{\nabla}\cdot\left(p\tilde{\nabla}u(\tilde{\mathbf{r}})\right)+qu(\tilde{\mathbf{r}})=0,\quad \text{for } \tilde{\mathbf{r}}\in\mathbf{\Omega}_{\mathrm{PML}} \tag{19}$$

which is explicitly written as shown in (3b). Here, $u(\tilde{\mathbf{r}})$ is the analytic continuation of the unknown function to complex space, and the nabla (or del) operator in complex space is expressed as

$$\tilde{\nabla}=(\mathbf{J}_c^{-1})^{\mathrm{T}}\cdot\nabla \tag{20}$$

where $\mathbf{J}_c$ is the Jacobian matrix of the transformation $(x,y)\to(\tilde{x},\tilde{y})$ defined in 2D Cartesian coordinates as

$$\mathbf{J}_c=\frac{\partial(\tilde{x},\tilde{y})}{\partial(x,y)}=\begin{bmatrix}\partial\tilde{x}/\partial x & \partial\tilde{x}/\partial y\\ \partial\tilde{y}/\partial x & \partial\tilde{y}/\partial y\end{bmatrix} \tag{21}$$

In FEM, the original form of the governing differential equation is not used, but instead converted to its weak variational form by using the weighted residual method. The weak form of the Helmholtz equation in (19) is given in complex space as

$$\int\limits_{\boldsymbol{\Omega}_{\rm PML}} p\tilde{\nabla}u(\tilde{\mathbf{r}})\cdot\tilde{\nabla}w(\tilde{\mathbf{r}})d\tilde{x}d\tilde{y}+\int\limits_{\boldsymbol{\Omega}_{\rm PML}} qu(\tilde{\mathbf{r}})w(\tilde{\mathbf{r}})d\tilde{x}d\tilde{y}=0 \tag{22}$$

where w is the weight function. The weak form is computed within each element after the computational domain is discretized using elements (such as triangular elements). The unknown function in each element is approximated as a linear combination of the shape (basis) functions weighted by coefficients that are the nodal values of the unknown function to be determined. That is,

$$u\left(\tilde{\mathbf{r}}\right)=\sum_{j=1}^{3}N_j\left(\tilde{\mathbf{r}}\right)u_j^e \tag{23}$$

where u_j^e is the j-th nodal value of the unknown function in the e-th element, and N_j is the shape (basis) function for the j-th node. The unknown function is substituted into the weak form in (22), and the weight functions are selected as the shape functions based on the Galerkin approach (i.e., $w=N_j$). Hence, the weak form becomes

$$\sum_{j=1}^{3}u_j^e\left[\iint\limits_{\boldsymbol{\Omega}^e_{\rm PML}} p^e\tilde{\nabla}N_i\cdot\tilde{\nabla}N_j d\tilde{x}d\tilde{y}+\iint\limits_{\boldsymbol{\Omega}^e_{\rm PML}} q^eN_iN_jd\tilde{x}d\tilde{y}\right]=0,\quad (i=1,2,3) \tag{24}$$

where the term in the square bracket is the ij-th entry of the 3×3 element matrix, which is explicitly written as

$$A_{ij}^e=\iint\limits_{\boldsymbol{\Omega}^e_{\rm PML}} p^e\left(\frac{\partial N_i}{\partial\tilde{x}}\frac{\partial N_j}{\partial\tilde{x}}+\frac{\partial N_i}{\partial\tilde{y}}\frac{\partial N_j}{\partial\tilde{y}}\right)d\tilde{x}d\tilde{y}+\iint\limits_{\boldsymbol{\Omega}^e_{\rm PML}} q^eN_iN_jd\tilde{x}d\tilde{y},\quad (i,j=1,2,3) \tag{25}$$

To ease the evaluation of the integrals in (25), the domain of the integration can be transformed to the master (ξ,η) domain using the isoparametric mapping approach. In the master domain, the representation of the unknown function given in (23) turns out to be

$$u\left(\xi,\eta\right)=\sum_{j=1}^{3}N_j\left(\xi,\eta\right)u_j^e \tag{26}$$

where the linear shape functions in the master domain are defined as

$$N_1(\xi,\eta)=1-\xi-\eta \tag{27a}$$

$$N_2(\xi,\eta)=\xi \tag{27b}$$

$$N_3(\xi,\eta)=\eta \tag{27c}$$

In fact, the mapping to the master domain introduces an additional coordinate transformation which transforms the complex coordinates to the master coordinates. A major difference between the LCPML and the LCPML-log methods lies on how these coordinate transformations are handled. The original LCPML utilizes the *complex space FEM* approach based on *complex elements* (i.e., elements with complex nodal coordinates), and hence the element matrix in (25) is computed using a transformation between complex and master coordinates, i.e., $(\xi,\eta)\to\left(\tilde{x},\tilde{y}\right)$ (see Fig. 5(a)). However, in the LCPML-log method, two successive coordinate transformations are used, one of which is the transformation between the master and real coordinates, and the other of which is the transformation between the real and complex coordinates, i.e., $(\xi,\eta)\to(x,y)\to\left(\tilde{x},\tilde{y}\right)$ (see Fig. 5(b)). In the original LCPML method, let us denote the Jacobian as $\mathbf{J}$ for the transformation $(\xi,\eta)\to\left(\tilde{x},\tilde{y}\right)$. In the LCPML-log method, the Jacobian of the cascaded transformation $(\xi,\eta)\to(x,y)\to\left(\tilde{x},\tilde{y}\right)$ can be computed as $\mathbf{J}=\mathbf{J}_m\mathbf{J}_c$ where $\mathbf{J}_m$ and $\mathbf{J}_c$ belong to the transformations $(\xi,\eta)\to(x,y)$ and $(x,y)\to\left(\tilde{x},\tilde{y}\right)$, respectively. Hence, the derivatives of the shape functions in (25) can be transformed to the master domain as follows:

$$\begin{bmatrix}\partial N_j/\partial\tilde{x}\\ \partial N_j/\partial\tilde{y}\end{bmatrix}=\mathbf{J}^{-1}\begin{bmatrix}\partial N_j/\partial\xi\\ \partial N_j/\partial\eta\end{bmatrix} \tag{28}$$

where $\mathbf{J}^{-1}$ is the inverse of the Jacobian matrix. In the LCPML-log method, the inverse of the Jacobian matrix is $\mathbf{J}^{-1}=\mathbf{J}_c^{-1}\mathbf{J}_m^{-1}$, where $\mathbf{J}_c^{-1}$ and $\mathbf{J}_m^{-1}$ are inverse of the Jacobian matrices $\mathbf{J}_c$ and $\mathbf{J}_m$, respectively. In addition, the differential area element $d\tilde{x}d\tilde{y}$ is replaced by $|\mathbf{J}|\,d\xi d\eta=|\mathbf{J}_m|\,|\mathbf{J}_c|\,d\xi d\eta$. Here, $|\cdot|$ denotes the determinant of the corresponding Jacobian matrix. As a result, the element matrix in (25) can be written as

$$\begin{aligned}A_{ij}^e=&\iint_{\boldsymbol{\Omega}^{\rm master}_{\rm PML}} p\left[\begin{array}{c}\left(J_{11}^{\rm inv}\frac{\partial N_i}{\partial\xi}+J_{12}^{\rm inv}\frac{\partial N_i}{\partial\eta}\right)\left(J_{11}^{\rm inv}\frac{\partial N_j}{\partial\xi}+J_{12}^{\rm inv}\frac{\partial N_j}{\partial\eta}\right)+\ldots\\ \ldots+\left(J_{21}^{\rm inv}\frac{\partial N_i}{\partial\xi}+J_{22}^{\rm inv}\frac{\partial N_i}{\partial\eta}\right)\left(J_{21}^{\rm inv}\frac{\partial N_j}{\partial\xi}+J_{22}^{\rm inv}\frac{\partial N_j}{\partial\eta}\right)\end{array}\right]|\mathbf{J}|\,d\xi d\eta\qquad(i,j=1,2,3)\\ &+\iint_{\boldsymbol{\Omega}^{\rm master}_{\rm PML}} qN_iN_j\,|\mathbf{J}|\,d\xi d\eta\end{aligned} \tag{29}$$

where $J_{ij}^{\rm inv}$ is the ij-th entry of the inverse Jacobian matrix. It is now worthwhile to note that the LCPML-log method is different from the original LCPML method in terms of how the Jacobian matrices are computed. The formulation details of the Jacobian matrices will be given in the subsections 4.1 and 4.2 below for the LCPML and LCPML-log methods, respectively.

4.1. Original LCPML method [16–18]

The original LCPML method is based on the complex FEM approach where elements with complex nodal coordinates are used to reflect the effect of the complex coordinate stretching defined in Sec. 3.1 on the propagating waves. Hence, this approach is very modular and easy to implement because the standard FEM formulation can be used without any modification, and the only action to be taken is to replace the real coordinates by their complex counterparts within the PML region. In the following, we briefly summarize how the Jacobian of the transformation is computed in the original LCPML method for the sake of completeness.

Using the transformation $(\xi,\eta)\rightarrow(\tilde{x},\tilde{y})$ and the *isoparametric mapping* approach, the coordinates are expressed in terms of the same shape functions, as given below (see Fig. 5(a)).

$$\tilde{x}=\sum_{j=1}^{3}N_j(\xi,\eta)\,\tilde{x}_j \quad \text{and}\, \tilde{y}=\sum_{j=1}^{3}N_j(\xi,\eta)\,\tilde{y}_j \tag{30}$$

where $\tilde{x}_j$ and $\tilde{y}_j$ are the *complex* coordinates of the j-th node in the complex triangular element. The Jacobian matrix of the transformation is obtained as

$$\mathbf{J}=\begin{bmatrix}\partial\tilde{x}/\partial\xi & \partial\tilde{y}/\partial\xi\\ \partial\tilde{x}/\partial\eta & \partial\tilde{y}/\partial\eta\end{bmatrix}=\begin{bmatrix}\tilde{x}_2-\tilde{x}_1 & \tilde{y}_2-\tilde{y}_1\\ \tilde{x}_3-\tilde{x}_1 & \tilde{y}_3-\tilde{y}_1\end{bmatrix} \tag{31}$$

It is clear from (31) that the Jacobian matrix depends only on the nodal complex coordinates. When linear triangular elements are used, the element matrix can be computed directly since the derivatives of the shape functions in (27) are *constant* and the inverse Jacobian matrix depends only on nodal coordinates which are also *constant*. As a result, the matrix entries are evaluated directly and the FEM is implemented in a standard way using complex-valued node coordinates rather than real-valued node coordinates.

Algorithm 1: Original LCPML method.

input: Nodal coordinate arrays (x_i, y_i) for $i=1,2,\ldots,N$
for i from 1 to N **do**
  if $(x_i, y_i)\in\mathbf{\Omega}_{\text{PML}}$ **then**
    Compute complex nodal coordinate arrays $(\tilde{x}_i,\tilde{y}_i)$ by scheme (8) and (10)-(16)
  else
    $\tilde{x}_i\leftarrow x_i$, $\tilde{y}_i\leftarrow y_i$
  end if
end for
return $(\tilde{x}_i,\tilde{y}_i)$ for $i=1,2,\ldots,N$
input: *conn* and $(\tilde{x}_i,\tilde{y}_i)$ for $i=1,2,\ldots,N$
for e from 1 to M **do** ▷ loop over all elements within PML region
  Calculate Jacobian $\mathbf{J}$ using (31)
  for g from 1 to n **do** ▷ loop over all Gauss points
    for i from 1 to 3 **do** ▷ construct 3×3 element matrix
      for j from 1 to 3 **do**
        Compute element matrix A^e_{ij} using (29)
      end for
    end for
  end for
  for i from 1 to 3 **do** ▷ assembly of global matrix
    $ii\leftarrow conn(e,i)$
    for j from 1 to 3 **do**
      $jj\leftarrow conn(e,j)$
      $A_{ii,jj}\leftarrow A_{ii,jj}+A^e_{ij}$
    end for
  end for
end for
return $A_{ii,jj}$ for $ii=1,2,\ldots,N$ and $jj=1,2,\ldots,N$

The calculation of the global matrix using the original LCPML is described by the Algorithm 1. In this algorithm, *conn* is the connectivity matrix, and M, N and n denote the number of elements within PML, the number of nodes and the number of Gauss points, respectively.

4.2. Proposed LCPML-log method

In the LCPML-log method, the derivative terms in the entries of the Jacobian matrix are computed numerically during the matrix formation phase of the FEM. In order to compute the element matrices in the master coordinates, two coordinate transformations are applied consecutively as follows: $(\xi,\eta)\rightarrow(x,y)\rightarrow(\tilde{x},\tilde{y})$ (see Fig. 5(b)). Hence, the complex coordinates can be expressed as composite functions shown below.

$$\tilde{x}(x,y)=\tilde{x}(x(\xi,\eta),y(\xi,\eta)) \quad \text{and} \quad \tilde{y}(x,y)=\tilde{y}(x(\xi,\eta),y(\xi,\eta)) \tag{32}$$

For the transformation $(\xi,\eta)\rightarrow(x,y)\rightarrow(\tilde{x},\tilde{y})$, we first express the real coordinates in terms of the shape functions as follows:

$$x=\sum_{j=1}^{3}N_j(\xi,\eta)\,x_j \quad \text{and} \quad y=\sum_{j=1}^{3}N_j(\xi,\eta)\,y_j \tag{33}$$

where x_j and y_j are the *real* coordinates of the j-th node in a triangular element.

By employing the chain rule, the derivatives of the shape functions can be expressed as follows:

$$\begin{bmatrix} \partial N_j/\partial\tilde{x} \\ \partial N_j/\partial\tilde{y} \end{bmatrix} = \mathbf{J}_c^{-1}\mathbf{J}_m^{-1} \begin{bmatrix} \partial N_j/\partial\xi \\ \partial N_j/\partial\eta \end{bmatrix} \tag{34}$$

where the Jacobian matrices $\mathbf{J}_m$ and $\mathbf{J}_c$, corresponding to the transformations $(\xi,\eta)\to(x,y)$ and $(x,y)\to(\tilde{x},\tilde{y})$ respectively, are expressed as

$$\mathbf{J}_m = \begin{bmatrix} \partial x/\partial\xi & \partial y/\partial\xi \\ \partial x/\partial\eta & \partial y/\partial\eta \end{bmatrix} = \begin{bmatrix} x_2-x_1 & y_2-y_1 \\ x_3-x_1 & y_3-y_1 \end{bmatrix} \tag{35}$$

$$\mathbf{J}_c = \begin{bmatrix} \partial\tilde{x}/\partial x & \partial\tilde{y}/\partial x \\ \partial\tilde{x}/\partial y & \partial\tilde{y}/\partial y \end{bmatrix} \tag{36}$$

As obvious from (35), the Jacobian matrix $\mathbf{J}_m$ depends only on the nodal real coordinates, which means that it is a constant matrix. However, the most critical part of the proposed LCPML-log method is to compute the entries of the Jacobian matrix $\mathbf{J}_c$ in (36), which are functions of the real coordinates inside the PML region. For this purpose, let us first consider the complex coordinates given by (8), where the unit vector, given in (12), is expressed as $\hat{\mathbf{n}} = n_x\hat{\mathbf{x}} + n_y\hat{\mathbf{y}}$. By employing the chain rule, the entries of the Jacobian matrix $\mathbf{J}_c$ become

$$\partial\tilde{x}/\partial x = 1 - jk^{-1}\left[n_x(1-\delta)^{-1}\partial\delta/\partial x + f\,\partial n_x/\partial x\right] \tag{37a}$$

$$\partial\tilde{x}/\partial y = -jk^{-1}\left[n_x(1-\delta)^{-1}\partial\delta/\partial y + f\,\partial n_x/\partial y\right] \tag{37b}$$

$$\partial\tilde{y}/\partial y = 1 - jk^{-1}\left[n_y(1-\delta)^{-1}\partial\delta/\partial y + f\,\partial n_y/\partial y\right] \tag{37c}$$

$$\partial\tilde{y}/\partial x = -jk^{-1}\left[n_y(1-\delta)^{-1}\partial\delta/\partial x + f\,\partial n_y/\partial x\right] \tag{37d}$$

Notice that the points on the inner and outer PML boundaries (i.e., $(x_{\text{in}},y_{\text{in}})$ and $(x_{\text{out}},y_{\text{out}})$) are functions of the corresponding PML point (x,y). Therefore, while differentiating the decay function and the unit vector components with respect to x and y, this function dependence must be included in the calculations. In the LCPML-log method, the differentiation is performed numerically to apply the method to more general geometries. Let us denote the function $v(x,y)$ as one of the functions δ, n_x or n_y. The derivatives can be computed by the 3-point central differencing formula obtained by Lagrange interpolation as follows:

$$\frac{\partial v(x,y)}{\partial x} \approx \frac{1}{2\Delta x}\left[v(x+\Delta x,y) - v(x-\Delta x,y)\right] \tag{38a}$$

$$\frac{\partial v(x,y)}{\partial y} \approx \frac{1}{2\Delta y}\left[v(x,y+\Delta y) - v(x,y-\Delta y)\right] \tag{38b}$$

The error in this approximation is $O\left(h^2\right)$, where h is the step size (Δx or Δy). Alternatively, the following 5-point differentiation formula can be used with error $O\left(h^4\right)$.

$$\frac{\partial v(x,y)}{\partial x} \approx \frac{1}{12\Delta x}\left[v(x-2\Delta x,y) - 8v(x-\Delta x,y) + 8v(x+\Delta x,y) - v(x+2\Delta x,y)\right] \tag{39a}$$

$$\frac{\partial v(x,y)}{\partial y} \approx \frac{1}{12\Delta y}\left[v(x,y-2\Delta y) - 8v(x,y-\Delta y) + 8v(x,y+\Delta y) - v(x,y+2\Delta y)\right] \tag{39b}$$

In order to compute the derivatives, the points on the inner and outer PML boundaries must be obtained. For some canonical geometries (such as circular and rectangular PML regions), these points can be obtained analytically. For the circular PML geometry where the inner and the outer radii of the PML region are a and b (see Fig. 6(a)), these points are defined as

$$x_{\text{in}} = \frac{ax}{\sqrt{x^2+y^2}} \quad \text{and} \quad y_{\text{in}} = \frac{ay}{\sqrt{x^2+y^2}} \tag{40}$$

$$x_{\text{out}} = \frac{bx}{\sqrt{x^2+y^2}} \quad \text{and} \quad y_{\text{out}} = \frac{by}{\sqrt{x^2+y^2}} \tag{41}$$

For the rectangular region (see Fig. 6(b)), the points are given in Table 1. For an arbitrary geometry, the points can be obtained numerically by searching among nodal coordinates of the PML boundaries defined in a typical FEM mesh using simple search techniques (e.g., using the *find* command in MATLAB). As a third alternative method, the boundary points can be obtained in a more systematic manner by using Lagrange multiplier method if the geometry of the PML boundary can be defined mathematically. This method is described in Sec. 4.2.1.

Since the entries of the Jacobian matrix $\mathbf{J}_c$ are functions of the real coordinates, the evaluation of the integrals in the element matrix is not very straightforward, compared to the original LCPML method. The integrals can be computed numerically by using the Gaussian quadrature rule. In this method, the integral of an arbitrary function $g(\xi,\eta)$ over the master triangle is computed by a weighted sum of function values at some specified points within the domain of integration, as given below.

$$\int_0^1\int_0^{1-\eta} g(\xi,\eta)\,d\xi d\eta \approx \sum_{k=1}^{n} w_k g(\xi_k,\eta_k) \tag{42}$$

where n is the number of Gauss points within the integration domain. When implementing the LCPML-log method, the Gauss points should be chosen so that they do not lie on the boundary of the triangular element connected to the outer PML boundary, due to the logarithmic singularity issue as discussed in Sec. 3.2. Hence, 4-point Gaussian quadrature, whose points and weights are referred to [20], can be used.

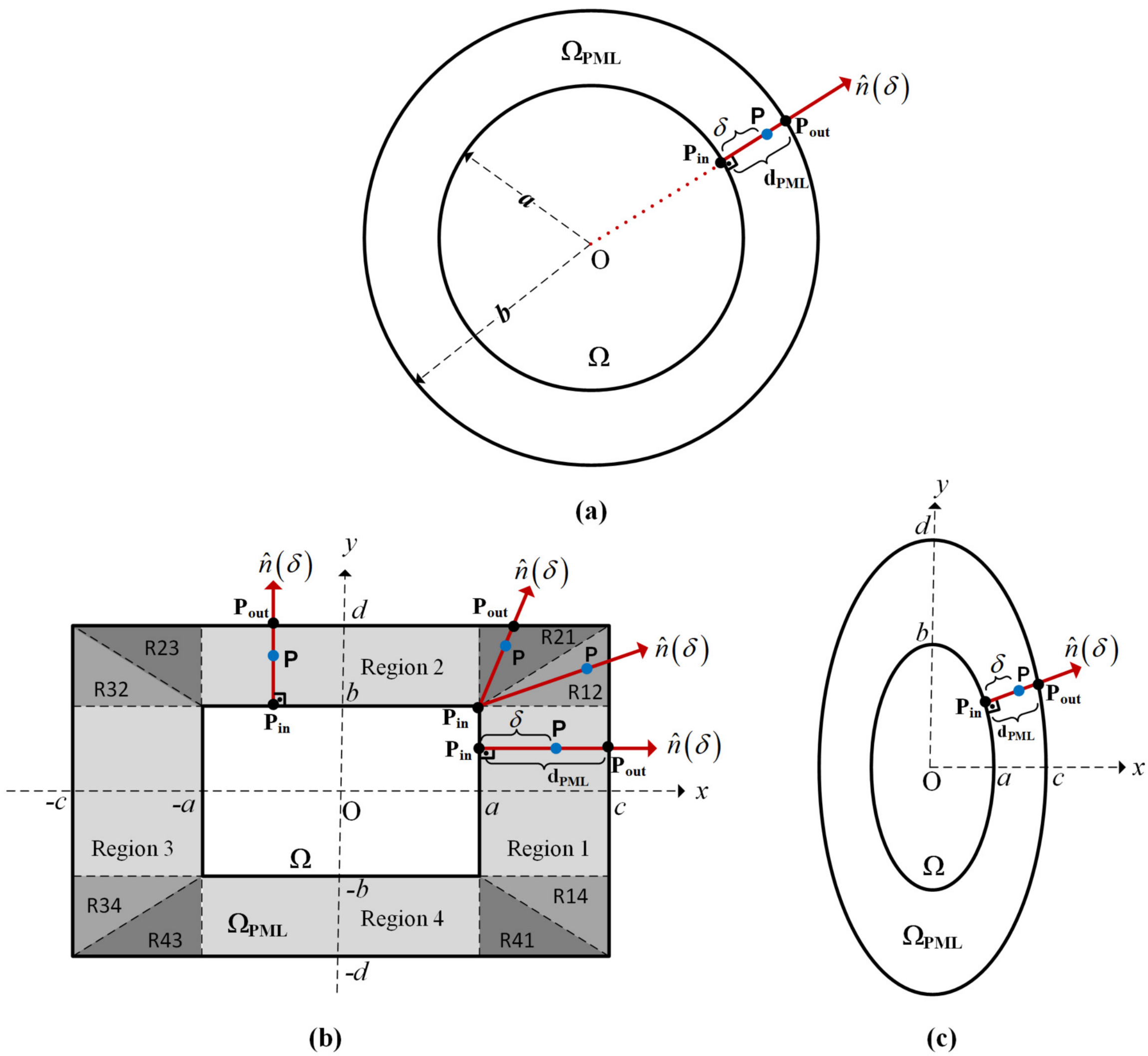

Fig. 6. The proposed LCPML-log method. (a) Circular region, (b) rectangular region, (b) elliptical region.

4.2.1. Determination of boundary points by Lagrange multiplier method

First, let us describe the basic principles of the Lagrange multiplier-based approach for finding the boundary points $(x_{\text{in}}, y_{\text{in}})$ on the inner PML boundary. Since these points must be determined with respect to the normal unit vector, they are in fact located at the shortest distance from the boundary. Therefore, we employ the Lagrange multiplier to find the minimum distance from the boundary to the corresponding PML point. Let us assume that the geometry of the inner PML boundary $(\partial\boldsymbol{\Omega}_{\text{in}})$ is represented by

$$g_{\text{in}}(x, y) = c \tag{43}$$

where c is a constant. We define a function, which represents the square of the distance between the PML point $(x_{\text{PML}}, y_{\text{PML}})$ and each point (x, y) on the boundary, as follows:

$$D(x, y, x_{\text{PML}}, y_{\text{PML}}) = (x - x_{\text{PML}})^2 + (y - y_{\text{PML}})^2 \tag{44}$$

The following condition must hold because the gradient vectors must be parallel to each other:

$$\nabla D = \lambda \nabla g_{\text{in}} \rightarrow \begin{bmatrix} 2(x - x_{\text{PML}}) \\ 2(y - y_{\text{PML}}) \end{bmatrix} = \lambda \begin{bmatrix} \partial g_{\text{in}}/\partial x \\ \partial g_{\text{in}}/\partial y \end{bmatrix} \tag{45}$$

where λ is the Lagrange multiplier. Note that ∇g_{in} is in the same direction as the normal unit vector representing the decay direction in the PML region. From (45), we get

$$x = x_{\text{PML}} + \frac{\lambda}{2}\frac{\partial g_{\text{in}}}{\partial x} \quad \text{and}\, y = y_{\text{PML}} + \frac{\lambda}{2}\frac{\partial g_{\text{in}}}{\partial y} \tag{46}$$

We denote x and y in (46) by the functions $x = \zeta(x_{\text{PML}}, y_{\text{PML}}, \lambda)$ and $y = \psi(x_{\text{PML}}, y_{\text{PML}}, \lambda)$, respectively. After substituting them into (43), for a given $(x_{\text{PML}}, y_{\text{PML}})$, we define

$$h(\lambda) = |g_{\text{in}}(\zeta(x_{\text{PML}}, y_{\text{PML}}, \lambda), \psi(x_{\text{PML}}, y_{\text{PML}}, \lambda)) - c| \tag{47}$$

Table 1
Definition of the points on the inner and outer PML boundaries for the rectangular PML region shown in Fig. 6(b) in the LCPML-log method.

Region 1 If (x >= a) & (y >= −b) & (y <= b)	**Region 2** If (y >= b) & (x >= −a) & (x <= a)	**Region 3** If (x <= −a) & (y >= −b) & (y <= b)	**Region 4** If (y <= −b) & (x >= −a) & (x <= a)
$x_{in}=a$ $y_{in}=y$ $x_{out}=c$ $y_{out}=y$	$x_{in}=x$ $y_{in}=b$ $x_{out}=x$ $y_{out}=d$	$x_{in}=-a$ $y_{in}=y$ $x_{out}=-c$ $y_{out}=y$	$x_{in}=x$ $y_{in}=-b$ $x_{out}=x$ $y_{out}=-d$
Region 12 If (x >= a) & (y >= b) & (y <= ((d-b)/(c-a))*(x-a)+b)	**Region 21** If (x >= a) & (y >= b) & (y > ((d-b)/(c-a))*(x-a)+b)	**Region 23** If (x <= −a) & (y >= b) & (y > ((d-b)/(−c+a))*(x+a)+b)	**Region 32** If (x <= −a) & (y >= b) & (y <= ((d-b)/(−c+a))*(x+a)+b)
$x_{in}=a$ $y_{in}=b$ $x_{out}=c$ $y_{out}=m\,(x_{out}-x)+y$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=a$ $y_{in}=b$ $y_{out}=d$ $x_{out}=(1/m)\,(y_{out}-y)+x$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=-a$ $y_{in}=b$ $y_{out}=d$ $x_{out}=(1/m)\,(y_{out}-y)+x$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=-a$ $y_{in}=b$ $x_{out}=-c$ $y_{out}=m\,(x_{out}-x)+y$ $m=(y-y_{in})\,/\,(x-x_{in})$
Region 34 If (x <= −a) & (y <= −b) & (y >= ((−d+b)/(−c+a))*(x+a)-b)	**Region 43** If (x <= −a) & (y <= −b) & (y < ((−d+b)/(−c+a))*(x+a)-b)	**Region 41** If (x >= a) & (y <= −b) & (y < ((−d+b)/(c-a))*(x-a)-b)	**Region 14** If (x >= a) & (y <= −b) & (y >= ((−d+b)/(c-a))*(x-a)-b)
$x_{in}=-a$ $y_{in}=-b$ $x_{out}=-c$ $y_{out}=m\,(x_{out}-x)+y$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=-a$ $y_{in}=-b$ $y_{out}=-d$ $x_{out}=(1/m)\,(y_{out}-y)+x$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=a$ $y_{in}=-b$ $y_{out}=-d$ $x_{out}=(1/m)\,(y_{out}-y)+x$ $m=(y-y_{in})\,/\,(x-x_{in})$	$x_{in}=a$ $y_{in}=-b$ $x_{out}=c$ $y_{out}=m\,(x_{out}-x)+y$ $m=(y-y_{in})\,/\,(x-x_{in})$

Hence, the problem can be formulated as an unconstrained optimization problem as follows:

$$\min_{\lambda} h\left(\lambda\right) \tag{48}$$

which can be solved for λ by using an optimization technique. For example, the function *fminsearch*, which is a derivative-free method used to find the minimum of unconstrained multivariable function in MATLAB, can be used to obtain λ that minimizes $h\left(\lambda\right)$. Once λ is computed, it is substituted into (46), which gives us the boundary points (x_{in}, y_{in}) on the inner PML boundary.

Second, the Lagrange multiplier-based approach can be used similarly to obtain the points (x_{out}, y_{out}) on the outer PML boundary. However, since these points must be determined by intersecting the outer boundary with the normal unit vector defined according to the inner boundary and the PML point, the Lagrange multiplier approach is slightly modified. Let us assume that the geometry of the outer PML boundary $(\partial\mathbf{\Omega}_{out})$ is represented by

$$g_{out}\left(x, y\right) = c \tag{49}$$

Since the vector passing through the PML point and the point on the inner boundary, which is in fact along the normal direction with respect the inner boundary, must be parallel to the vector passing through the points on the inner and outer boundaries (or alternatively the vector passing through the PML point and the point on the outer boundary), we express

$$\begin{bmatrix} x - x_{in} \\ y - y_{in} \end{bmatrix} = \lambda \begin{bmatrix} x_{PML} - x_{in} \\ y_{PML} - y_{in} \end{bmatrix} \tag{50}$$

which results in

$$x = \lambda x_{PML} + x_{in}\left(1-\lambda\right) \quad \text{and} \quad y = \lambda y_{PML} + y_{in}\left(1-\lambda\right) \tag{51}$$

We similarly write (51) as $x = \zeta\left(x_{PML}, y_{PML}, x_{in}, y_{in}, \lambda\right)$ and $y = \psi\left(x_{PML}, y_{PML}, x_{in}, y_{in}, \lambda\right)$, and substitute them into (49). For a given (x_{PML}, y_{PML}) and (x_{in}, y_{in}), the problem is formulated as a minimization problem, similar to (48), where

$$h\left(\lambda\right) = \left|g_{out}\left(\zeta\left(x_{PML}, y_{PML}, x_{in}, y_{in}, \lambda\right), \psi\left(x_{PML}, y_{PML}, x_{in}, y_{in}, \lambda\right)\right) - c\right| \tag{52}$$

After λ is computed by solving the minimization problem, (x_{out}, y_{out}) is determined by (51).

Let us now apply this approach to elliptical PML region shown in Fig. 6(c). The equation of the inner PML boundary $(\partial\mathbf{\Omega}_{in})$ is written as $x^2/a^2 + y^2/b^2 = 1$. Hence, (46) becomes

$$x = \frac{x_{PML}}{1-\lambda/a^2} \quad \text{and} \quad y = \frac{y_{PML}}{1-\lambda/b^2} \tag{53}$$

and (47) is expressed as

$$h\left(\lambda\right) = \left|\frac{a^2 x_{PML}^2}{\left(a^2-\lambda\right)^2} + \frac{b^2 y_{PML}^2}{\left(b^2-\lambda\right)^2} - 1\right| \tag{54}$$

Afterwards, the equation of the outer PML boundary ($\partial\mathbf{\Omega}_{\text{out}}$) is written as $x^2/c^2 + y^2/d^2 = 1$.

We express (52) as

$$h(\lambda) = \left|\left[\lambda x_{\text{PML}} + x_{\text{in}}(1-\lambda)\right]^2/c^2 + \left[\lambda y_{\text{PML}} + y_{\text{in}}(1-\lambda)\right]^2/d^2 - 1\right| \tag{55}$$

The formation of the global matrix within the PML region by using the LCPML-log method is described by the Algorithm 2. The algorithms show the procedure when the elements are within the PML region. For elements outside the PML region, the standard FEM algorithm is used.

Algorithm 2: Proposed LCPML-log method.

input: *conn* and (x_i, y_i) for $i = 1, 2, ..., N$
for e from 1 to M **do** ▷ loop over all elements within PML region
  Calculate Jacobian $\mathbf{J}_m$ using (39)
  for g from 1 to n **do** ▷ loop over all Gauss points
    $x \leftarrow \sum_{j=1}^{3} N_j(\xi_g, \eta_g)\, x_j$ ▷ eqn. (33)
    $y \leftarrow \sum_{j=1}^{3} N_j(\xi_g, \eta_g)\, y_j$ ▷ eqn. (33)
    Calculate Jacobian $\mathbf{J}_c$ using (36)-(41) and Sec. 4.2.1
    $\mathbf{J} \leftarrow \mathbf{J}_m \mathbf{J}_c$
  for i from 1 to 3 **do** ▷ construct 3×3 element matrix
    for j from 1 to 3 **do**
      Compute element matrix A_{ij}^e using (29)
    end for
  end for
  end for
  for i from 1 to 3 **do** ▷ assembly of global matrix
    $ii \leftarrow conn(e, i)$
    for j from 1 to 3 **do**
      $jj \leftarrow conn(e, j)$
      $A_{ii,jj} \leftarrow A_{ii,jj} + A_{ij}^e$
    end for
  end for
end for
return $A_{ii,jj}$ for $ii = 1, 2, ..., N$ and $jj = 1, 2, ..., N$

It is worthwhile to add this remark here is that the general presentation of the proposed LCPML-log method can also be used for other element types (such as quadrilateral elements) in a similar manner, except that the shape functions and the master element of the corresponding element type must be employed in the FEM formulation as is done in the standard FEM formulation. The formulation peculiar to the proposed LCPML-log method, especially eqn. (35)-(41) and Sec. 4.2.1, is also valid for other element types.

5. Numerical examples

In this section, the accuracy of the LCPML-log method is tested by comparing the results with those of the original LCPML method, the classical PML method and the analytical results. To conduct a systematic analysis, we consider the problem where the free-space Green's function for Helmholtz equation is constructed in a given domain. The problem is formulated by replacing the excitation function in (1a) by the Dirac delta (or impulse) function as follows:

$$\nabla \cdot (p\nabla u) + qu = -\delta(\mathbf{r} - \mathbf{r}_s), \tag{56}$$

where $\mathbf{r}_s$ is the location of a line source inside $\mathbf{\Omega}$. The analytical solution of (56), which is indeed the free-space Green's function, is

$$u(\mathbf{r}) = (j/4)\, H_0^{(2)}(k\,|\mathbf{r} - \mathbf{r}_s|) \tag{57}$$

where $H_0^{(2)}$ is the Hankel function of the second kind of zeroth order. This problem provides a convenient way of measuring the accuracy of the proposed LCPML-log method because it forms a basis for general radiation and scattering problems. Any excitation (or source) function can be represented as a convolution integral due to the linearity and space-invariance property of the problems governed by the Helmholtz equation. That is,

$$f_e(\mathbf{r}) = -\int\limits_{\mathbf{\Omega}_f} f_e(\mathbf{r}_s)\, \delta(\mathbf{r} - \mathbf{r}_s)\, d\mathbf{r}_s \tag{58}$$

where $\mathbf{\Omega}_f = \operatorname{supp}(f_e)$. Since FEM is a field-based method rather than source-based method, the FEM solution of (56) can be achieved by converting the problem into a boundary value problem governed by the homogeneous Helmholtz equation with a Dirichlet type boundary condition (BC) as follows:

$$\nabla \cdot (p\nabla u) + qu = 0 \tag{59a}$$

$$\text{with BC: } u = u_a \text{ on } \partial\mathbf{\Omega}_s \tag{59b}$$

Table 2
MSE values for the simulations in Fig. 7.

Wavelength (λ)	PML type	MSE		
		2 layers	3 layers	5 layers
$\lambda = 1$ m	Classical PML	3.82e-3 (min at $\alpha_c = 4$)	3.07e-3 (min at $\alpha_c = 3$)	2.78e-3 (min at $\alpha_c = 2$)
	LCPML with α_c ($m = 2$)	3.28e-3 (min at $\alpha_c = 8$)	2.81e-3 (min at $\alpha_c = 5$)	2.64e-3 (min at $\alpha_c = 3$)
	LCPML with α_c ($m = 3$)	4.33e-3 (min at $\alpha_c = 10$)	2.93e-3 (min at $\alpha_c = 8$)	2.61e-3 (min at $\alpha_c = 5$)
	LCPML-log	5.35e-5	3.73e-5	3.21e-5
$\lambda = 4$ m	Classical PML	6.86e-3 (min at $\alpha_c = 15$)	2.58e-4 (min at $\alpha_c = 11$)	7.06e-5 (min at $\alpha_c = 8$)
	LCPML with α_c ($m = 2$)	6.51e-4 (min at $\alpha_c = 30$)	1.69e-4 (min at $\alpha_c = 24$)	3.67e-5 (min at $\alpha_c = 17$)
	LCPML with α_c ($m = 3$)	1.44e-3 (min at $\alpha_c = 38$)	3.89e-4 (min at $\alpha_c = 33$)	5.87e-5 (min at $\alpha_c = 25$)
	LCPML-log	5.78e-4	1.25e-4	2.08e-5
$\lambda = 16$ m	Classical PML	3.65e-2 (min at $\alpha_c = 38$)	1.09e-3 (min at $\alpha_c = 30$)	2.66e-4 (min at $\alpha_c = 22$)
	LCPML with α_c ($m = 2$)	5.13e-3 (min at $\alpha_c = 122$)	1.65e-3 (min at $\alpha_c = 67$)	2.98e-4 (min at $\alpha_c = 51$)
	LCPML with α_c ($m = 3$)	1.61e-2 (min at $\alpha_c = 90$)	3.59e-3 (min at $\alpha_c = 114$)	6.79e-4 (min at $\alpha_c = 76$)
	LCPML-log	6.56e-3	1.45e-3	2.61e-4

where u_a is the analytical solution given in (57), and $\partial\mathbf{\Omega}_s$ is the boundary of a 'small' region around the source location. In other words, a small region is defined around the source location and the analytical fields that are imposed along the boundary of this region are used as excitation fields in the FEM.

The results are compared in terms of the MSE (relative root-mean-square error), defined as

$$MSE = \sum_{i=1}^{N} \frac{\left|u^{(i)} - u_a^{(i)}\right|^2}{\left|u_a^{(i)}\right|^2} \tag{60}$$

where u and u_a are the computed and analytical solutions obtained at the nodes of the triangular mesh in $\mathbf{\Omega}$, and N is the number of the corresponding nodes.

In the following, five analyses are performed, as summarized below.

Analysis 1: The effect of the PML parameter variations in the classical PML and the original LCPML methods is analyzed. Both accuracy and robustness of the solutions are compared with the LCPML-log method.

Analysis 2: The effect of the number of PML layers is analyzed by plotting the *MSE vs. number of PML layers* graphs for different wavelength (λ) values.

Analysis 3: The effect of the wavelength (or frequency) is shown by plotting the *MSE vs. wavelength* graphs for different number of PML layers.

Analysis 4: The effect of source location is analyzed by taking 400 source points distributed uniformly within the computational domain. FEM is run for each source position and the results are aggregated to obtain a statistical analysis of MSE values by source location. The simulations are performed for each method and for different number of PML layers.

Analysis 5: The condition number is analyzed for different wavelength values and different number of PML layers.

Analysis 1 (Effect of parameter variations in the classical PML and the original LCPML methods):
As mentioned in Sec. 2 and 3, the performances of the classical PML and the original LCPML methods depend on the choice of PML parameters (i.e., $\alpha = \alpha_c k$ and PML thickness). To illustrate this behavior, let us now consider a square domain whose half-length is $a = 2$ m. The region is excited by a line source located at a point as shown in Fig. 7(a). In the generated mesh, the element size and the thickness of each layer are 0.05 m. The same mesh is used in all simulations in this section. The MSE values are obtained as a function of the α_c parameter for different wavelength values and different number of PML layers, and plotted in Fig. 7 for both classical PML and the original LCPML methods. In these plots, logarithmic axes are used for better visualization. As shown in these plots, there is an optimal value of α_c for minimizing the MSE value, but this optimal value is frequency dependent. Therefore, the α_c parameter should adaptively be assigned during the frequency sweep, which is obviously a challenging task due to the difficulty of obtaining an optimal value for different problems. The difficulty of assigning the α_c parameter in the original LCPML method can be overcome and the original LCPML can be implemented by defining the coordinate transformation as proposed in (16). Here, the 'small' parameter ϵ corresponds to the magnitude of the wave on the outer PML boundary. Practically, this parameter can be chosen as $\epsilon = 10^{-2}$, as long as a sufficient number of PML layers (for example, more than 3 layers) is used. The performance of the original LCPML method, proposed in (16), as a function of the ϵ parameter is shown in Fig. 8. It is worthwhile to note that the same problem is simulated by the LCPML-log method without the need for any PML parameter tuning. In Table 2, we compare the MSE values of the classical PML, the original LCPML and the proposed LCPML-log methods. The results show that the LCPML-log method usually gives more accurate results, even if the parameters of the classical and the original LCPML methods are adjusted for best performance. Hence, we conclude that the LCPML-log method gives more accurate and robust results especially for small (1 to 3) number of PML layers.

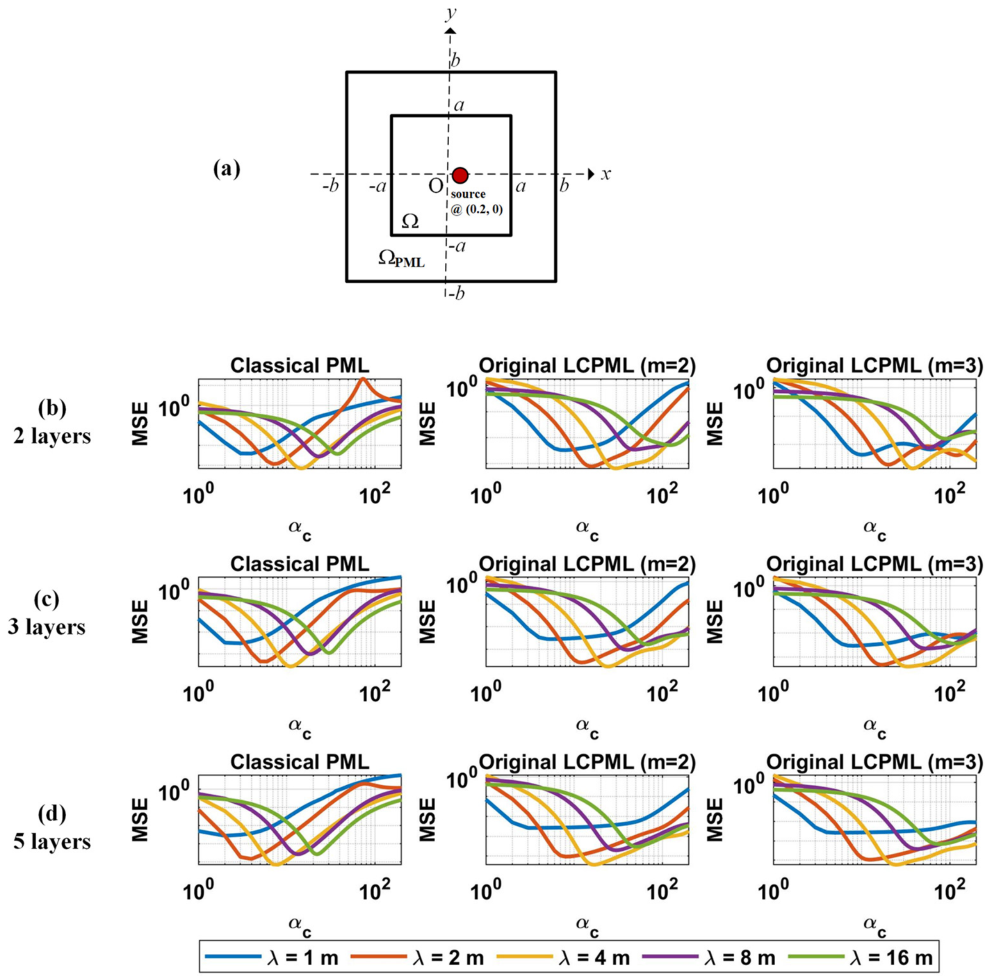

Fig. 7. Analysis 1: MSE versus α_c parameter in the classical PML and the original LCPML methods. (a) Square region, (b) MSE for 2 layers, (c) MSE for 3 layers, (d) MSE for 5 layers (*axes are logarithmic*).

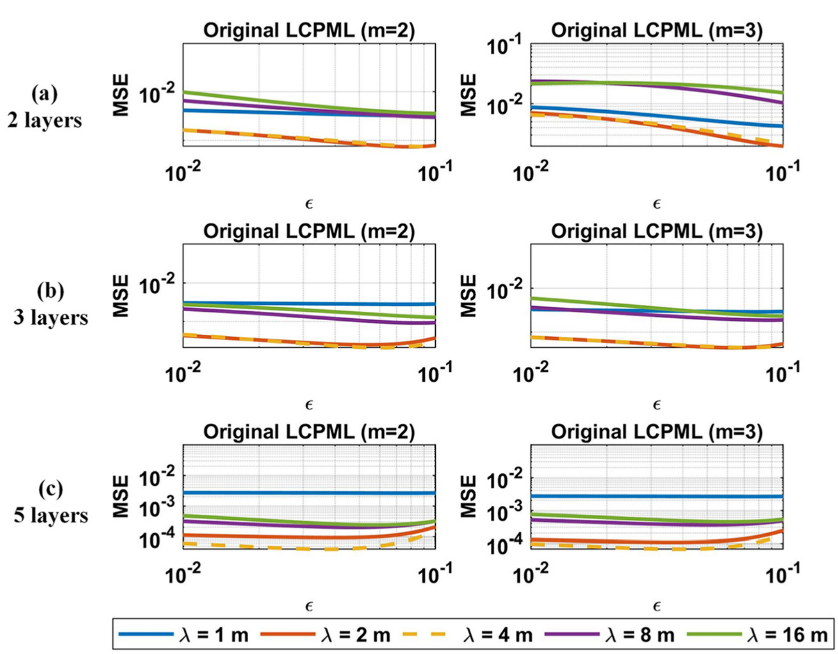

Fig. 8. Analysis 1: MSE versus ϵ parameter in the original LCPML method. (a) 2 layers, (b) 3 layers, (c) 5 layers (*axes are logarithmic*).

Table 3
Analysis parameters for the simulations in Fig. 9.

Wavelength (λ) (m)	1	2	4
Frequency (MHz)	300	150	75
Element size and the thickness of each layer (0.05 m)	$\lambda/20$	$\lambda/40$	$\lambda/80$
Circle radius *ka* ($a = 1$ m)	2π	π	$\pi\ /\ 2$
Ellipse semi-minor and semi-major axes (*ka*, *kb*) ($a = 1$ m) ($b = 2$ m)	$(2\pi, 4\pi)$	$(\pi, 2\pi)$	$(\pi/2, \pi)$
Rectangle half-lengths (*ka*, *kb*) ($a = 2$ m) ($b = 1$ m)	$(4\pi, 2\pi)$	$(2\pi, \pi)$	$(\pi, \pi/2)$

Table 4
Analysis parameters for the simulations in Figs. 10 and 12.

Wavelength Interval (m) $\lambda_{\min} \le \lambda \le \lambda_{\max}$	$\lambda_{\min} = 2$ m	$\lambda_{\max} = 30$ m
Frequency Interval (MHz) $f_{\max} \ge f \ge f_{\min}$	$f_{\max} = 150$ MHz	$f_{\min} = 10$ MHz
Element size and the thickness of each layer (0.05 m)	$\lambda/40$ (*max*)	$\lambda/600$ (*min*)
Circle radius *ka* ($a = 3$ m)	3π (*max*)	0.2π (*min*)
Ellipse semi-minor and semi-major axes (*ka*, *kb*) ($a = 3$ m) ($b = 6$ m)	$(3\pi, 6\pi)$ (*max*)	$(0.2\pi, 0.4\pi)$ (*min*)
Rectangle half-lengths (*ka*, *kb*) ($a = 6$ m) ($b = 3$ m)	$(6\pi, 3\pi)$ (*max*)	$(0.4\pi, 0.2\pi)$ (*min*)

Analysis 2 (MSE vs. Number of PML Layers):
We compare the performances of the LCPML and LCPML-log methods by changing the number of PML layers. In this and subsequent analyses, the LCPML, defined in (16) where $\epsilon = 10^{-2}$, is considered. For each of the circular, elliptical and rectangular domains, mesh is created with the given dimensions shown in Table 3 and Fig. 9. A line source is located at a point shown in Fig. 9. The MSE values are obtained for each wavelength, and the results are plotted in a comparative manner. As shown, the performance of the LCPML-log method is better than the original LCPML method especially when a few PML layers are used. The MSE value tends to increase as the wavelength decrease because the element size increases in terms of wavelength. Hence, this expected difference is mainly due the mesh resolution rather than the performance of the methods.

Analysis 3 (MSE vs. Wavelength):
We present the MSE results as a function of wavelength (or frequency), as shown in Fig. 10. The dimensions are shown in Table 4. The purpose of using the same mesh size when the frequency/wavelength changes is that many applications require the simulation of a structure by conducting a frequency sweep over a broader band of frequencies with the same mesh. Such a simulation possesses challenges in terms of both computational load and the accuracy that must be satisfied with different mesh resolutions in this frequency band. As is evident from the flatness of the graphs, the performance of the LCPML-log method is almost independent of the frequency of operation even when a relatively coarser mesh is used.

Analysis 4 (MSE vs. Source Location):
We analyze the effect of the source location on the performance of the methods. We select 400 source locations which are uniformly distributed within $\boldsymbol{\Omega}$ and run FEM for each source position to obtain a family MSE results. Afterwards, scattering plots, histogram graphs and some statistical parameters (such as mean and variance) are obtained for each family of the results. The results are shown in Fig. 11 for the circular, elliptical and rectangular regions at $\lambda = 2$ m. It is observed that the LCPML-log method provides robust and reliable results regardless of the source location even a single layer is used. The accuracy slightly decreases when the source is close to the inner PML boundary and to the corners or sharp regions of the geometry.

Analysis 5 (Condition Number):
In the last analysis, we compare the condition numbers of the matrices arising in the original LCPML and the LCPML-log methods. The results are plotted in Fig. 12 by considering the analysis parameters in Table 4. It is observed that the LCPML-log method improves the condition number compared to the LCPML method, whose condition number is indeed parameter-dependent.

Finally, the classical PML, the original LCPML and the LCPML-log methods are *qualitatively* compared in Table 5 according to some criteria.

6. Conclusions

In this paper, we have introduced the parametrization-free LCPML-log method using a new coordinate transformation based on a logarithmic decay function. We have shown through various numerical results that the proposed method provides accurate and robust results with a few PML layers. We have observed that the error values obtained by the proposed method in the simulations are usually less than 1e-2 or even smaller, even if a single layer is used. In addition, it does not require any structured mesh inside the PML region, as opposed to the methods in [9,10], and can be applied to arbitrary mesh structure created using conventional mesh generation methods. It is also worth noting that the LCPML-log method can easily be applied to other types of waves, such as acoustical waves, because of the similarity between the governing wave equations and the Helmholtz equation and the exact duality between the parameters of interest. The proposed method will be extended to the FEM solution of complex domains, such as those involving layered inhomogeneous media, and three-dimensional electromagnetic radiation and scattering problems, and its performance will be further investigated in the near future.

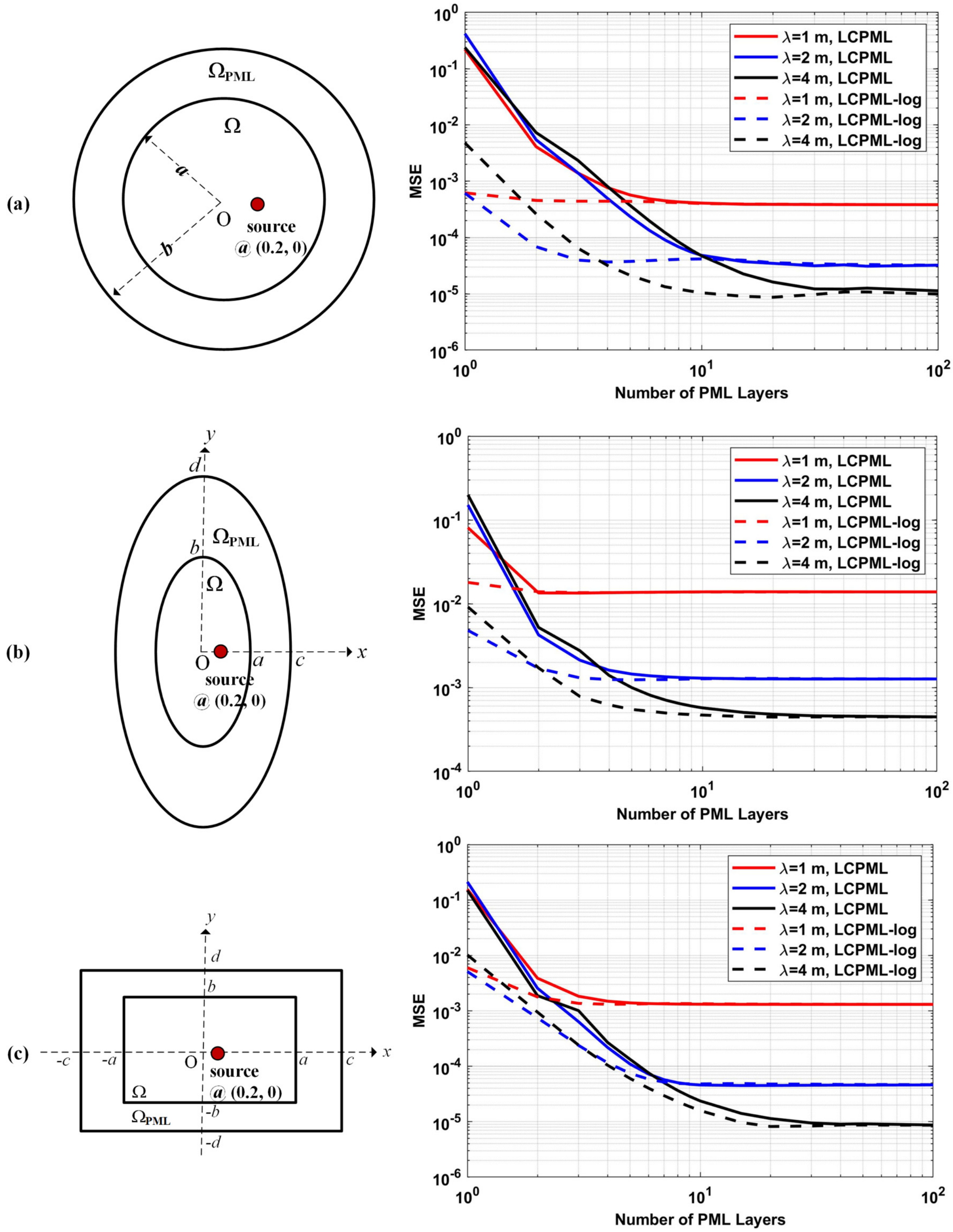

Fig. 9. Analysis 2: MSE vs. number of PML layers. (a) Circular region, (b) elliptical region, (c) rectangular region (*axes are logarithmic*).

The developed MATLAB codes, which are used to perform the numerical examples in this paper, are available at Appendix A so that the readers can benefit from the LCPML methods.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

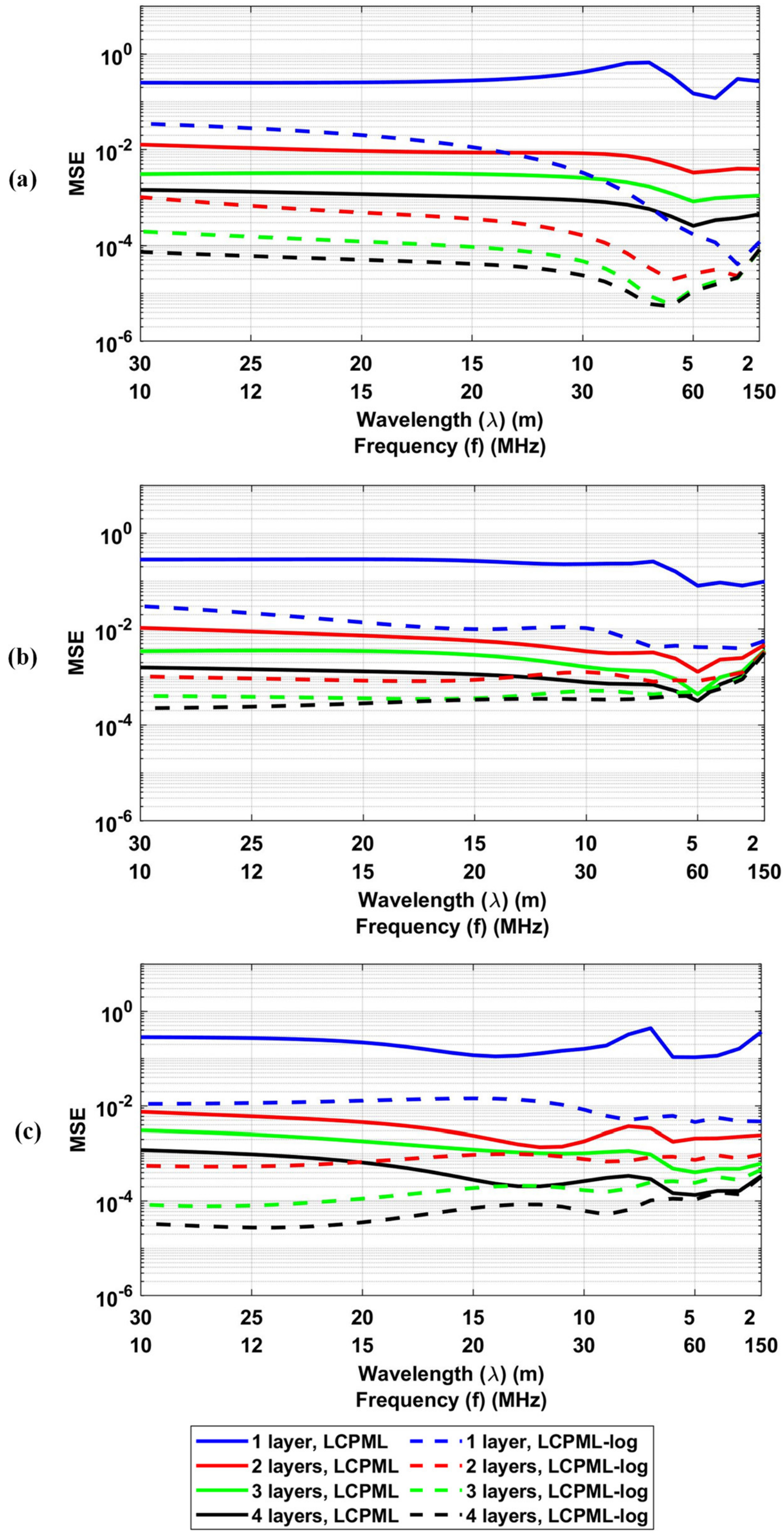

Fig. 10. Analysis 3: MSE vs. wavelength. (a) Circular region, (b) elliptical region, (c) rectangular region (*y-axis is logarithmic*).

Data availability

The developed MATLAB codes are available as a supplementary material.

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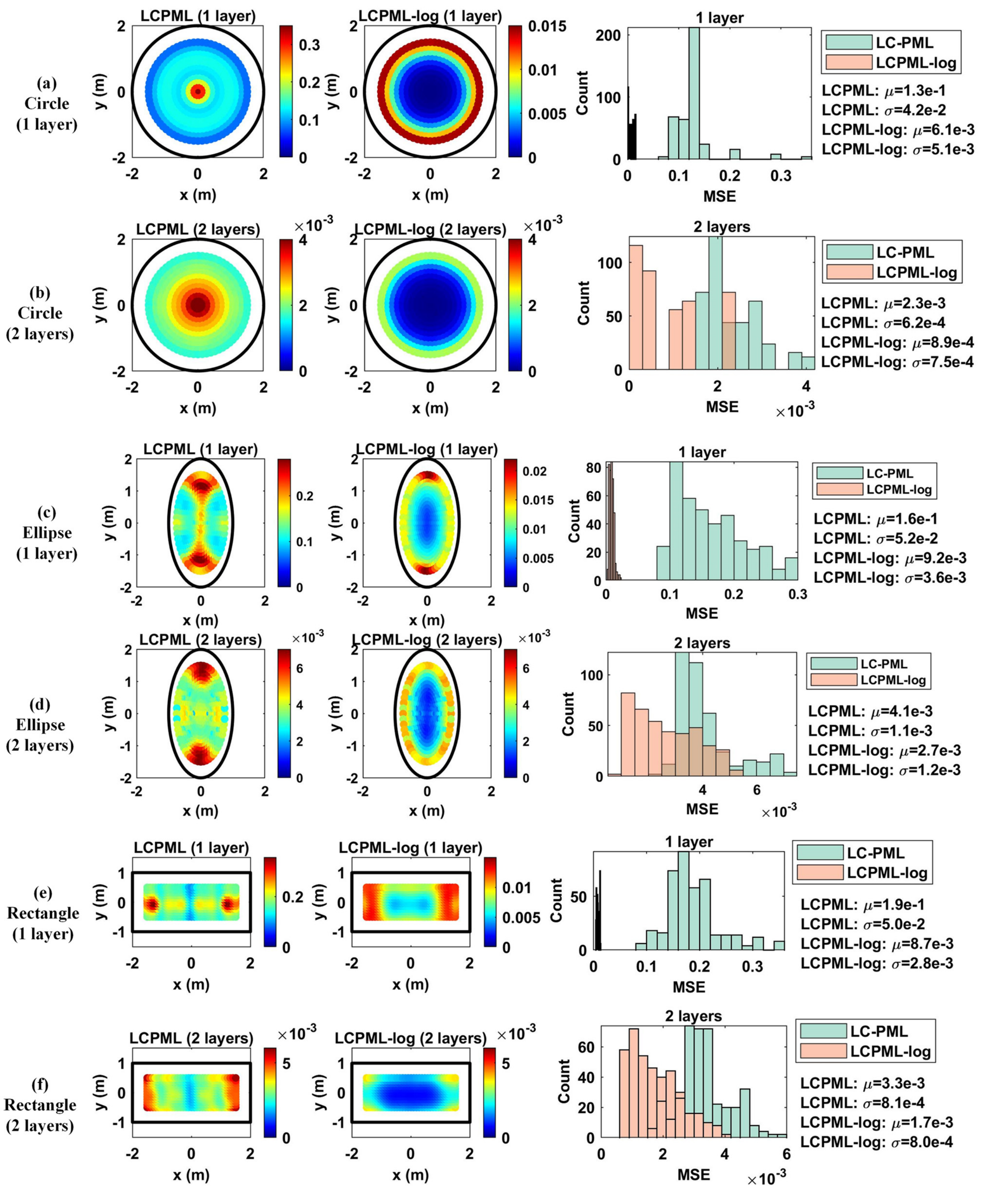

Fig. 11. Analysis 4: Scattering plots and histograms of MSE vs. source location. (a) Circle (1 layer), (b) circle (2 layers), (c) ellipse (1 layer), (d) ellipse (2 layers), (e) rectangle (1 layer), (f) rectangle (2 layers) (μ: mean, σ: standard deviation).

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cpc.2023.108741.

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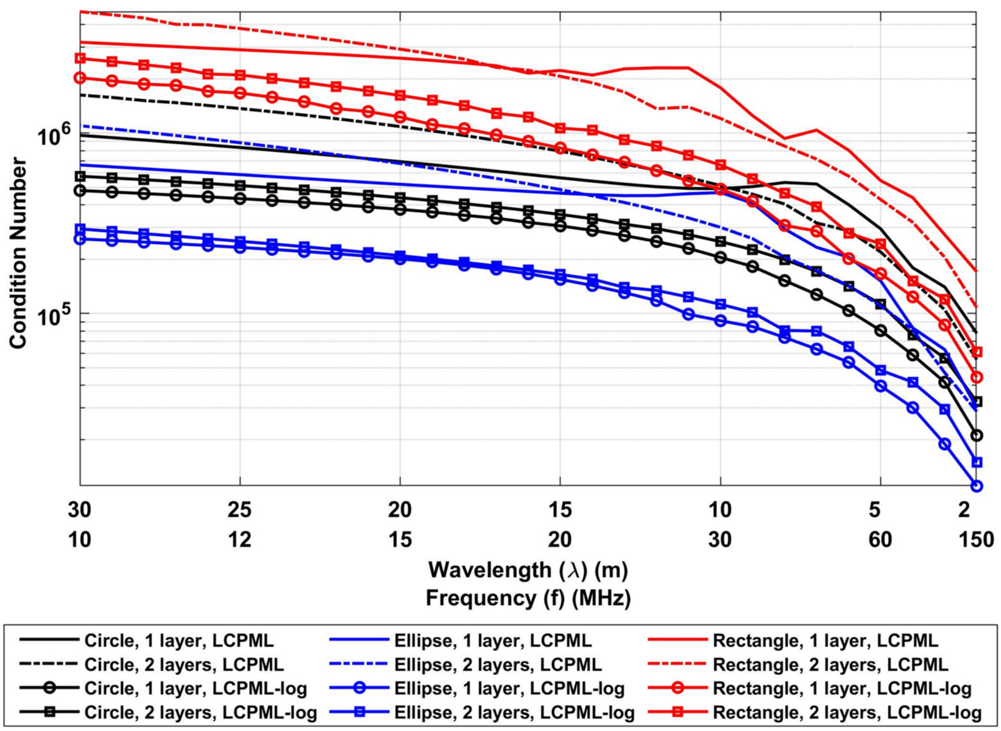

Fig. 12. Analysis 5: Condition number vs. wavelength (*y-axis is logarithmic*).

Table 5
Qualitative comparison of the classical PML, original LCPML and LCPML-log methods.

Criterion	Classical PML	Original LCPML (with α_c parameter)	Original LCPML (with ϵ parameter)	Proposed LCPML-log
Applicability to arbitrary convex domains (easy/hard if applicable)	No	Yes (easy)	Yes (easy)	Yes (easy)
Robustness (in terms of PML parameter variations with small number (1 to 3) of PML layers)	Low	Low	Medium	High
Accuracy with small number (1 to 3) of PML layers	Medium	Medium	Medium	High
Modularity of PML implementation in a FEM code	High	High	High	High
Computation time	Low	Low	Low	Medium

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